Knowledge Management Systems and the Activation of Dormant Data

Abstract

Organizations today operate within digitally saturated environments in which enterprise systems, supply chains, platforms, and customer interfaces continuously generate extensive data trails. Yet a substantial share of these digital traces remains unexamined, poorly classified, or disconnected from strategic decision processes. This underutilized informational residue—often characterized as dormant or dark data—represents both a governance liability and a latent source of competitive advantage. While prior research has established the performance implications of IT investments and digital capabilities, limited attention has been given to the systematic role of Knowledge Management Systems (KMS) in transforming dormant data into actionable organizational knowledge.

This article develops a conceptual framework that positions KMS as activation mechanisms embedded within digital infrastructures. Building on the IT business value perspective (Melville, Kraemer, & Gurbaxani, 2004) and capability-based views of digital assets (Aral & Weill, 2007), the analysis argues that dormant data does not yield performance benefits absent complementary processes for structuring, contextualizing, and integrating information into organizational routines. Drawing on the notion of digital options and agility (Sambamurthy, Bharadwaj, & Grover, 2003), we conceptualize activated data as a portfolio of strategic options that enhances responsiveness and innovation capacity.

The framework further integrates research on digital infrastructures and generativity (Tilson et al., 2010; Henfridsson & Bygstad, 2013), suggesting that KMS enhance infrastructural evolution by rendering dispersed data discoverable, interoperable, and recombinable. Insights from digital innovation scholarship (Yoo, Henfridsson, & Lyytinen, 2010; Nambisan et al., 2017; Kohli & Melville, 2019) inform the argument that data activation is not merely a technical exercise but a socio-technical process involving knowledge codification, cross-boundary collaboration, and innovation orchestration.

We propose a multi-stage activation model encompassing identification, structuring, contextualization, integration, and recombination. This process reframes dormant data from passive storage artifacts into generative knowledge resources capable of driving service innovation, supply chain optimization, and strategic agility. The article contributes by articulating the mechanisms through which KMS convert latent data accumulation into measurable organizational value, thereby extending the digital innovation and IT performance literature toward a more knowledge-centric understanding of data-driven transformation.

Keywords

Knowledge Management Systems; Dormant Data; Digital Infrastructure; Digital Innovation; Organizational Agility; Dynamic Capabilities

1. Introduction

The contemporary organization operates within a dense ecology of digital systems that continuously capture, transmit, and store data. Enterprise resource planning platforms, customer relationship management systems, supply chain networks, collaboration tools, and cloud-based applications collectively generate vast digital traces of operational activity. These traces include transactional records, system logs, communication metadata, sensor outputs, archival documents, and analytic by-products. While the rhetoric of digital transformation emphasizes real-time analytics and data-driven decision-making, empirical reality reveals a persistent gap between data accumulation and knowledge utilization. A substantial portion of organizational data remains unstructured, unanalyzed, or disconnected from strategic processes. This accumulation of underexploited informational assets—often referred to as dormant or dark data—raises fundamental questions about how firms can systematically convert digital abundance into sustained organizational value.

Research on IT business value provides an important starting point for understanding this challenge. Foundational work demonstrates that investments in IT infrastructure do not automatically translate into performance gains; rather, value emerges when technological resources are complemented by organizational capabilities and process redesign (Melville, Kraemer, & Gurbaxani, 2004). Similarly, evidence indicates that IT assets enhance performance when they are embedded within coordinated organizational routines and governance structures (Aral & Weill, 2007). These insights imply that dormant data is not inherently valuable; its strategic potential depends on the presence of mechanisms that render it interpretable, accessible, and actionable.

At the same time, the evolution of digital technologies has altered the logic of organizing. Digital artifacts possess unique characteristics—reprogrammability, data homogeneity, and layered modular architectures—that enable continuous recombination and innovation (Yoo, Henfridsson, & Lyytinen, 2010). Digital infrastructures expand and evolve through generative mechanisms that integrate new actors, standards, and functionalities (Tilson et al., 2010; Henfridsson & Bygstad, 2013). Within such environments, data is not merely an output of operations but a constitutive element of innovation ecosystems. Yet generativity presupposes visibility and interoperability. Data that remains buried in silos, legacy systems, or poorly governed repositories cannot participate in recombinatorial innovation processes.

The paradox of digital transformation thus lies in simultaneous data abundance and knowledge scarcity. Organizations invest heavily in analytics platforms and digital initiatives, yet often overlook the structured processes required to activate historical and residual data assets. Recent scholarship on dark data highlights the economic, strategic, and risk implications of underutilized data resources (Olaitan et al., 2025). Dormant data may conceal insights into customer behavior, operational inefficiencies, or emerging market opportunities. Conversely, it may pose compliance and cybersecurity risks when left unmanaged. The strategic question is not simply how to collect more data, but how to systematically convert accumulated digital artifacts into institutionalized knowledge that informs decision-making and innovation.

This article argues that Knowledge Management Systems (KMS) provide the critical socio-technical infrastructure for activating dormant data. Traditionally conceptualized as repositories of documents, lessons learned, or expert directories, KMS have often been treated as peripheral to core digital strategy. However, within digitally intensive organizations, KMS can function as integrative platforms that bridge data analytics, organizational memory, and collaborative interpretation. By embedding metadata standards, taxonomies, knowledge codification processes, and cross-functional interfaces, KMS enable firms to transform raw data into structured knowledge objects.

The strategic significance of this transformation becomes clearer when viewed through the lens of digital options and agility. The concept of digital options describes IT-enabled capabilities that expand a firm's strategic flexibility and responsiveness (Sambamurthy, Bharadwaj, & Grover, 2003). Activated dormant data constitutes a portfolio of latent strategic options—insights that can be rapidly mobilized in response to environmental shifts. For example, archived customer interaction logs may reveal emerging demand patterns; historical supply chain data may uncover resilience vulnerabilities; legacy R&D documentation may inform product recombination. Without activation mechanisms, such informational assets remain inert. With effective KMS integration, they become components of a dynamic capability system that enhances agility.

Digitally enabled supply chains provide a concrete illustration of this logic. Research demonstrates that performance improvements arise when information integration enhances coordination, visibility, and decision quality across supply networks (Rai, Patnayakuni, & Seth, 2006). Yet supply chain systems generate extensive historical and operational data that often remains underutilized once immediate transactions are completed. By integrating dormant data into knowledge repositories and analytics workflows, firms can identify recurring bottlenecks, supplier performance patterns, and demand volatility signals. Thus, KMS-mediated activation extends the benefits of digital integration beyond real-time coordination to long-term learning and innovation.

The activation of dormant data must also be understood within the broader context of digital innovation management. Digital innovation involves the recombination of digital and physical components to create new products, services, or business models (Nambisan et al., 2017). This recombination depends on accessible knowledge resources and interoperable infrastructures. Service innovation in digitally mediated contexts relies heavily on data-driven insights that inform customer experience design and value proposition development (Barrett et al., 2015). Dormant data represents an underexploited reservoir of such insights. However, its contribution to innovation requires institutional mechanisms that facilitate discovery, contextualization, and cross-boundary collaboration.

Despite the recognized importance of knowledge processes, existing literature often treats data analytics and knowledge management as distinct domains. Analytics research emphasizes algorithmic processing and predictive modeling, whereas knowledge management scholarship focuses on codification, sharing, and organizational learning. The activation of dormant data requires integrating these perspectives. Data must be algorithmically processed, but it must also be socially interpreted, validated, and embedded within decision routines. KMS serve as the

integrative layer that connects technical analytics capabilities with organizational learning processes.

Moreover, digital infrastructures are characterized by heterogeneity and layered complexity. Legacy systems coexist with cloud platforms; structured databases coexist with unstructured communication archives. This heterogeneity contributes to the proliferation of dormant data, as disparate systems lack standardized metadata and interoperability. Research on infrastructure evolution highlights the importance of generative mechanisms that enable adaptation and extension (Henfridsson & Bygstad, 2013). KMS can enhance infrastructural generativity by providing shared ontologies, classification schemas, and knowledge governance protocols that make dormant data interoperable across layers.

The stakes of dormant data activation extend beyond efficiency gains. In an era of heightened environmental uncertainty and rapid technological change, organizational resilience depends on the capacity to leverage historical knowledge and latent insights. Digital innovation review research underscores that competitive advantage increasingly derives from the orchestration of digital resources rather than isolated technological assets (Kohli & Melville, 2019). Dormant data activation through KMS contributes to this orchestration by aligning dispersed informational assets with strategic objectives.

Importantly, activation is not a purely technical endeavor. It involves governance structures, cultural norms, and incentive systems that encourage knowledge reuse and cross-functional collaboration. IT business value research emphasizes complementarities between technological investments and organizational processes (Melville et al., 2004). Without supportive governance and leadership commitment, KMS risk becoming static repositories rather than dynamic capability platforms. Thus, activating dormant data requires integrating technological design with managerial strategy.

The present study addresses three interrelated research gaps. First, while dormant data has been identified as a growing phenomenon, there is limited theoretical articulation of the mechanisms through which it can be systematically activated. Second, although KMS are widely implemented, their role in mediating between data accumulation and digital innovation outcomes remains underexplored. Third, existing frameworks often emphasize either infrastructure evolution or capability development, but seldom integrate these perspectives into a coherent model of data activation.

To address these gaps, this article develops a process-oriented conceptual framework that links dormant data activation to organizational agility, innovation capacity, and performance outcomes. The framework synthesizes insights from IT business value research (Melville et al., 2004; Aral & Weill, 2007), digital options theory (Sambamurthy et al., 2003), supply chain integration (Rai et al., 2006), digital infrastructure evolution (Tilson et al., 2010; Henfridsson & Bygstad, 2013), and digital innovation management (Yoo et al., 2010; Nambisan et al., 2017; Kohli & Melville, 2019).

Conceptually, we argue that dormant data activation unfolds through interconnected stages: identification, structuring, contextualization, integration, and recombination.

Identification involves auditing and mapping dispersed data assets. Structuring entails cleaning, categorizing, and applying metadata standards. Contextualization embeds data within organizational narratives and process linkages. Integration connects knowledge objects to decision systems and workflows. Recombination leverages activated knowledge for innovation and strategic renewal. KMS operate as the institutional backbone supporting each stage, providing the technological and social infrastructure for sustained activation.

By articulating this framework, the article reframes dormant data from a passive by-product of digital operations into a strategic reservoir of digital options. It highlights the importance of knowledge infrastructures in shaping innovation trajectories and organizational performance. In doing so, it contributes to ongoing conversations about digital transformation, moving beyond analytics hype toward a deeper understanding of how organizations can systematically convert data abundance into enduring knowledge-based advantage.

The remainder of the article elaborates this framework, discusses organizational implications, and proposes directions for empirical research. Through this analysis, we aim to advance a knowledge-centric perspective on digital transformation—one in which Knowledge Management Systems are not peripheral repositories but central engines for activating dormant data and unlocking its generative potential.

2. Methodology

2.1 Research Design

This study adopts a theory-building, integrative conceptual methodology aimed at developing a process model of dormant data activation through Knowledge Management Systems (KMS). Rather than relying on a single empirical dataset, the research synthesizes and integrates established theoretical streams in information systems, digital innovation, and IT business value to construct a mechanism-level explanation of how dormant data can be systematically transformed into actionable organizational knowledge.

The methodological approach follows a structured theory synthesis design. This design is appropriate where fragmented literatures address related phenomena but lack integration around a central construct—in this case, dormant data activation. The study therefore combines deductive reasoning from established theoretical frameworks with abductive conceptual development to articulate a coherent activation model. The objective is explanatory rather than purely descriptive: to clarify causal pathways linking data accumulation, knowledge infrastructures, and organizational outcomes such as agility and innovation performance.

2.2 Theoretical Sampling and Literature Selection

The literature base was selected using theoretical sampling principles. Rather than conducting a broad systematic review across all knowledge management research, the selection focused on foundational and high-impact contributions that directly inform the mechanisms underlying dormant data activation.

Five theoretical domains guided selection:

1. **IT Business Value and Complementarity** – to ground the relationship between technological assets and organizational performance (Melville, Kraemer, & Gurbaxani, 2004; Aral & Weill, 2007).
2. **Digital Options and Organizational Agility** – to conceptualize activated data as a source of strategic flexibility (Sambamurthy, Bharadwaj, & Grover, 2003).
3. **Digitally Enabled Supply Chains and Information Integration** – to illustrate how data integration improves coordination and performance (Rai, Patnayakuni, & Seth, 2006).
4. **Digital Infrastructure and Generativity** – to explain how digital systems evolve and enable recombination (Tilson et al., 2010; Henfridsson & Bygstad, 2013).
5. **Digital Innovation and Service Transformation** – to link activated knowledge assets to innovation outcomes (Yoo, Henfridsson, & Lyytinen, 2010; Nambisan et al., 2017; Barrett et al., 2015; Kohli & Melville, 2019).

Additionally, recent work on dark data and value creation potential (Olaitan et al., 2025) was incorporated to define the phenomenon of dormant data and establish its contemporary relevance.

Selection criteria included:

- Theoretical rigor and citation impact.
- Clear articulation of mechanisms linking digital resources to performance.
- Conceptual relevance to data integration, knowledge processes, or digital innovation.
- Complementarity across domains to enable cross-theoretical synthesis.

This focused theoretical sampling ensured depth of conceptual integration while maintaining analytical coherence.

2.3 Conceptual Development Procedure

The development of the activation framework followed a three-phase analytical procedure:

Phase 1: Construct Clarification

Key constructs were defined and differentiated to prevent conceptual ambiguity.

- **Dormant Data** was conceptualized as digitally stored information assets that are technically accessible but not actively integrated into decision-making or innovation processes.
- **Knowledge Management Systems (KMS)** were defined as socio-technical infrastructures combining repositories, taxonomies, metadata governance, collaborative tools, and integration interfaces.
- **Activation** was conceptualized as a transformation process through which dormant data becomes structured, contextualized, and embedded within organizational routines.

This clarification phase ensured consistent construct boundaries and minimized overlap between data analytics, knowledge management, and infrastructure concepts.

Phase 2: Mechanism Extraction

From each theoretical stream, mechanism-level insights were extracted. For example:

- From IT business value research: the importance of complementarities between technology and organizational processes.
- From digital options theory: the idea that IT capabilities create future strategic flexibility.
- From infrastructure theory: generative mechanisms that enable recombination and expansion.
- From digital innovation literature: orchestration and modular recombination as drivers of new value creation.

Each mechanism was translated into a provisional activation function (e.g., "metadata standardization enhances interoperability," "knowledge codification reduces interpretive ambiguity," "integration increases recombinability").

Phase 3: Process Modeling

The extracted mechanisms were organized into a staged process model. Rather than presenting activation as a linear pipeline, the model conceptualizes interconnected stages: identification, structuring, contextualization, integration, and recombination.

Each stage was linked to supporting theoretical foundations and to anticipated organizational outcomes. The process logic emphasizes cumulative capability development: earlier stages enhance the effectiveness of later ones, while feedback loops enable continuous refinement.

Multi-Stage Dormant Data Activation Process

Activation Stage	Core Activities	Enabling KMS Mechanisms	Expected Intermediate Outcomes
Identification	Data auditing; asset mapping; metadata tagging	Data catalogues; discovery tools	Increased visibility of dormant datasets
Structuring	Cleaning; classification; taxonomy alignment	Ontologies; metadata governance frameworks	Improved interoperability
Contextualization	Annotation; narrative linking; process mapping	Knowledge repositories; collaboration platforms	Enhanced interpretability
Integration	Workflow embedding; dashboard linkage	API integration; analytics connectors	Improved decision integration

Recombination	Innovation experimentation; cross-domain synthesis	Cross-functional collaboration tools; innovation labs	Expanded innovation potential

2.4 Analytical Logic and Validation

The validity of the conceptual model was assessed through theoretical triangulation. Each stage of the activation process was evaluated against multiple theoretical lenses to ensure internal consistency and explanatory robustness. For example:

- The **identification stage** aligns with governance and capability perspectives emphasizing visibility and asset awareness.
- The **structuring stage** reflects infrastructure and modularity principles, ensuring interoperability.
- The **contextualization stage** corresponds to knowledge codification and organizational learning principles.
- The **integration stage** aligns with supply chain integration and digital coordination research.
- The **recombination stage** draws on digital innovation and generativity theory.

By ensuring that each stage is supported by multiple theoretical foundations, the framework avoids dependence on a single explanatory paradigm.

Additionally, boundary conditions were articulated to clarify the model's applicability. The framework is most relevant in digitally intensive organizations with heterogeneous infrastructures and significant data accumulation. It may be less applicable in organizations with minimal digital footprints or highly centralized data governance structures.

2.5 Operationalization Pathways for Future Empirical Research

Although the present study is conceptual, it provides clear guidance for empirical operationalization. Future research can measure:

- **Dormant Data Intensity:** proportion of stored datasets not linked to active workflows.
- **KMS Maturity:** presence of metadata governance, taxonomy depth, collaboration tools, and analytics integration.
- **Activation Effectiveness:** frequency of knowledge reuse, cross-functional data integration metrics, and innovation recombination rates.
- **Organizational Outcomes:** agility indicators, innovation outputs, supply chain performance metrics, and financial performance measures.

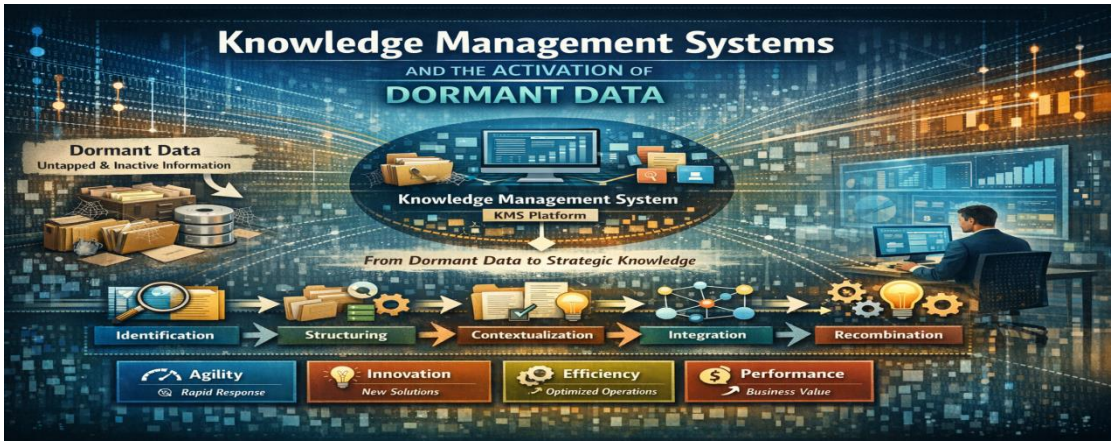
Multi-method empirical strategies could include longitudinal case studies, survey-based structural equation modeling, or archival performance analysis. The staged model lends itself to path analysis, enabling researchers to test mediation effects between KMS capabilities and performance outcomes.

2.6 Methodological Contribution

This methodological approach contributes by demonstrating how integrative theory synthesis can clarify emerging digital phenomena that span multiple research domains. Rather than isolating dormant data as a narrow technical issue, the methodology embeds it within broader conversations about infrastructure evolution, digital capabilities, and innovation ecosystems.

By combining deductive theoretical integration with process modeling, the study advances a structured explanation of dormant data activation. The result is a robust, mechanism-based framework that can guide both empirical testing and managerial application.

In sum, the methodology provides a rigorous conceptual foundation for understanding how Knowledge Management Systems transform dormant data into strategic knowledge assets, thereby bridging the gap between data accumulation and digital innovation performance.



3. Results

The results of this study are presented in the form of a theoretically derived activation model that explains how Knowledge Management Systems (KMS) convert dormant data into measurable organizational capabilities and performance outcomes. Because the research adopts a conceptual synthesis methodology, the results are explanatory rather than statistical. They demonstrate how the integration of IT business value theory, digital infrastructure research, and digital innovation management yields a coherent mechanism linking dormant data activation to agility, innovation, and performance improvements.

3.1 Dormant Data as a Latent Strategic Resource

The first key result is the reconceptualization of dormant data as a latent strategic resource rather than merely excess storage or governance overhead. Drawing from complementarity theory in IT business value research (Melville, Kraemer, & Gurbaxani, 2004; Aral & Weill, 2007), the findings indicate that dormant data possesses unrealized performance potential contingent on the presence of complementary organizational capabilities. Without structured activation mechanisms,

dormant data remains inert. However, when embedded within a mature KMS environment, it becomes a source of cumulative learning and strategic insight.

The model clarifies that dormant data contributes to performance only when three conditions are met:

1. Discoverability through metadata and classification.
2. Interpretability through contextualization and codification.
3. Integrability into workflows and decision systems.

These conditions form the structural preconditions for activation.

3.2 Multi-Stage Activation as a Capability Development Process

The second major result is the identification of a five-stage activation pathway—identification, structuring, contextualization, integration, and recombination—through which dormant data is progressively transformed into organizational knowledge assets.

Identification increases visibility of dispersed datasets, reducing informational fragmentation across digital infrastructures. This stage addresses the infrastructural opacity highlighted in digital infrastructure research (Tilson et al., 2010).

Structuring introduces standardized taxonomies and metadata schemas that enhance interoperability across heterogeneous systems. This aligns with generative infrastructure theory (Henfridsson & Bygstad, 2013), demonstrating that structured knowledge artifacts are more easily recombined and extended.

Contextualization embeds meaning into structured data through annotation, narrative linkage, and process alignment. This stage reflects knowledge codification mechanisms necessary for reducing interpretive ambiguity across functional boundaries.

Integration connects contextualized knowledge assets with operational systems, analytics dashboards, and decision routines. Evidence from digitally enabled supply chains (Rai, Patnayakuni, & Seth, 2006) suggests that such integration improves coordination quality and operational responsiveness.

Recombination leverages activated knowledge assets for innovation and service redesign, consistent with digital innovation theory (Yoo, Henfridsson, & Lyytinen, 2010; Nambisan et al., 2017). Activated data becomes recombinable digital components within broader innovation ecosystems.

Importantly, the results show that activation is cumulative: earlier stages enhance the efficiency and impact of later stages. The model therefore conceptualizes activation as a capability-building trajectory rather than a one-time data cleaning initiative.

3.3 Enhancement of Digital Agility

A third key result is the link between dormant data activation and organizational agility. Drawing on digital options theory (Sambamurthy, Bharadwaj, & Grover,

2003), the findings demonstrate that activated knowledge assets expand a firm's portfolio of digital options. Historical datasets, once structured and integrated, can be rapidly mobilized in response to market volatility or operational disruptions.

For example, archived operational logs may reveal recurring bottlenecks that inform rapid contingency planning. Historical customer interaction data may uncover shifting preference patterns that guide agile product adjustments. The activation process therefore increases both sensing and responding capabilities, core dimensions of digital agility.

The model predicts that organizations with mature KMS infrastructures will exhibit faster adaptation cycles and more effective strategic pivots than those with fragmented data environments.

3.4 Strengthening of Innovation Capacity

A fourth result concerns digital innovation capacity. The recombination stage demonstrates how activated knowledge resources contribute to service innovation and new value proposition development. Digital innovation management research emphasizes modular recombination as a central driver of value creation (Nambisan et al., 2017; Barrett et al., 2015). The results show that dormant data, once activated, becomes a reusable innovation input.

Activated datasets can be combined with emerging technologies such as AI analytics, automation tools, or platform services to generate new offerings. For instance, structured historical service data can support predictive maintenance services, while integrated customer archives can enable personalized digital experiences.

Thus, KMS-mediated activation enhances generativity within digital infrastructures by expanding the set of available recombinable components.

3.5 Performance Implications

The fifth major result is the articulation of a mediated pathway from KMS maturity to organizational performance. Rather than asserting a direct causal link, the model specifies a chain of mediation:

KMS Capabilities → Dormant Data Activation → Digital Agility and Innovation → Organizational Performance

This pathway aligns with IT business value research emphasizing indirect performance effects (Melville et al., 2004). The results indicate that performance gains are contingent upon successful activation and capability translation.

Expected performance outcomes include:

- Improved operational efficiency through learning from historical data patterns.
- Enhanced supply chain coordination via integrated knowledge flows.
- Increased innovation output due to expanded recombination capacity.
- Strengthened strategic resilience in volatile environments.

Mediated Pathway from KMS to Organizational Performance

Level	Construct	Description	Theoretical Basis
Level 1	KMS Capabilities	Metadata governance; collaboration; integration tools	IT complementarity theory
Level 2	Dormant Data Activation	Visibility; interpretability; interoperability; recombinability	Infrastructure & knowledge theory
Level 3	Dynamic Capabilities	Agility; sensing; responsiveness; innovation capacity	Digital options & innovation theory
Level 4	Organizational Outcomes	Operational efficiency; service innovation; resilience; performance	IT business value literature

3.6 Governance and Risk Reduction Effects

An additional result concerns governance and risk management. Activation processes improve transparency and oversight of accumulated data assets, reducing regulatory and cybersecurity vulnerabilities associated with unmanaged dormant datasets (Olaitan et al., 2025). Structured metadata governance enhances compliance monitoring and data lifecycle management.

Thus, dormant data activation contributes not only to opportunity creation but also to risk mitigation.

3.7 Emergent Dynamic Capability

Finally, the results suggest that KMS-enabled activation functions as a higher-order dynamic capability. Over time, organizations that continuously identify and reactivate dormant data develop institutional routines for knowledge renewal. This ongoing process strengthens the adaptability of digital infrastructures and enhances long-term competitiveness (Kohli & Melville, 2019).

Rather than treating dormant data activation as a discrete project, the model positions it as an embedded organizational capability that evolves alongside digital infrastructure expansion.

4. Discussion

This study advances the conversation on digital transformation by repositioning dormant data from a peripheral storage concern to a central strategic asset contingent on Knowledge Management System (KMS) capabilities. The discussion elaborates the theoretical implications, clarifies conceptual contributions, and outlines broader organizational insights that emerge from the activation framework.

4.1 Theoretical Implications

First, the findings extend IT business value research by specifying a previously underarticulated intermediate mechanism: dormant data activation. Prior work demonstrates that IT investments produce performance gains when complemented by organizational capabilities (Melville, Kraemer, & Gurbaxani, 2004; Aral & Weill, 2007). However, much of that literature focuses on operational systems or analytics in active use. The present framework clarifies that substantial digital value may reside in already-collected but underutilized data assets. Thus, performance effects are not only a function of new technology deployment but also of how organizations reactivate accumulated informational residues.

Second, the model refines the concept of digital options and agility (Sambamurthy, Bharadwaj, & Grover, 2003). Digital options are typically associated with forward-looking IT flexibility. This study suggests that dormant data, once activated, expands a firm's portfolio of options retrospectively as well as prospectively. Historical datasets become reservoirs of insight that can be mobilized during environmental shocks, regulatory shifts, or market realignments. Agility, therefore, is not only about rapid sensing of new data streams but also about intelligent reuse of archived knowledge.

Third, the framework contributes to digital infrastructure theory. Research on digital infrastructures emphasizes generativity, modularity, and evolution (Tilson et al., 2010; Henfridsson & Bygstad, 2013). Yet generativity depends on accessible, interoperable components. Dormant data activation enhances infrastructural generativity by converting opaque archives into modular, recombinable knowledge objects. KMS serve as mediating layers that reduce fragmentation and enable cross-layer recombination. In this sense, the study connects knowledge governance with infrastructure evolution, highlighting how socio-technical design influences innovation capacity.

Fourth, the discussion enriches digital innovation management literature (Yoo, Henfridsson, & Lyytinen, 2010; Nambisan et al., 2017; Kohli & Melville, 2019). Innovation scholarship frequently emphasizes recombination of digital components but tends to focus on real-time or newly generated data. The present model expands the innovation lens to include previously neglected data assets. Dormant data activation increases the diversity of recombinable components within digital ecosystems, thereby strengthening innovation optionality and service transformation potential (Barrett et al., 2015).

4.2 Conceptual Clarifications

The discussion also clarifies that activation is not synonymous with analytics. While advanced analytics tools may process dormant datasets, activation requires additional steps: metadata governance, contextual interpretation, integration into routines, and organizational sensemaking. Analytics without knowledge codification risks generating isolated insights that fail to diffuse across the enterprise. KMS provide the socio-technical infrastructure necessary to institutionalize insights and embed them within collective memory.

Furthermore, the staged activation model underscores the cumulative nature of capability development. Organizations that skip foundational stages—such as structured identification and metadata standardization—may struggle with interoperability and recombination later. The framework therefore challenges purely technology-centric approaches to digital transformation and instead emphasizes disciplined knowledge architecture.

4.3 Organizational and Strategic Insights

From a strategic standpoint, the discussion suggests that dormant data activation can function as a cost-effective pathway to value creation. Unlike large-scale system replacements or new digital platform investments, activation leverages already existing digital assets. This reframing is particularly relevant in resource-constrained contexts where maximizing the utility of accumulated data may yield substantial returns without additional data acquisition costs.

The governance dimension is equally important. Dormant data can pose compliance, cybersecurity, and privacy risks if unmanaged (Olaitan et al., 2025). Activation processes introduce visibility, lifecycle management, and accountability structures that mitigate such risks. Thus, KMS-driven activation aligns value creation with risk reduction, strengthening organizational resilience.

The discussion also highlights cultural implications. Effective activation depends on cross-functional collaboration and incentives for knowledge sharing. Organizational silos often perpetuate data dormancy by limiting interpretive exchange. KMS must therefore be embedded within supportive cultural norms that encourage reuse, annotation, and collaborative sensemaking.

4.4 Boundary Conditions and Limitations

The conceptual model is most applicable in digitally intensive organizations with heterogeneous infrastructures and substantial archival data accumulation. Firms with minimal digital footprints may experience limited activation potential. Additionally, activation outcomes depend on governance maturity and leadership commitment. Without strategic alignment, KMS risk becoming passive repositories rather than dynamic capability platforms.

4.5 Future Research Directions

The discussion points toward several research avenues. Empirical studies can test the mediation pathway linking KMS maturity, activation effectiveness, agility, and performance outcomes. Longitudinal research could examine how activation capabilities evolve over time alongside infrastructure expansion. Comparative studies across industries may reveal sector-specific activation patterns and governance challenges.

Further research might also explore the interaction between artificial intelligence systems and KMS in accelerating dormant data activation. As AI-driven tools become more prevalent, understanding how automated analytics integrates with human-centered knowledge codification will be critical.

4.6 Concluding Perspective

In summary, the discussion reinforces the central claim that Knowledge Management Systems are not peripheral repositories but strategic infrastructures that unlock latent digital value. By articulating the mechanisms through which dormant data becomes institutionalized knowledge, this study bridges fragmented literatures and provides a coherent explanation of how data abundance can translate into sustained innovation, agility, and performance advantage in digitally evolving organizations.

5. Conclusion

This article has advanced a knowledge-centric perspective on digital transformation by demonstrating that the strategic value of data does not reside solely in real-time analytics or newly generated digital streams, but also in the systematic activation of dormant data. Across digitally intensive organizations, vast repositories of underutilized data accumulate as by-products of routine operations, supply chain transactions, customer interactions, and legacy systems. Without structured activation mechanisms, these data assets remain inert—incurring storage and governance costs while contributing little to organizational learning or innovation.

By integrating insights from IT business value research (Melville, Kraemer, & Gurbaxani, 2004; Aral & Weill, 2007), digital options theory (Sambamurthy, Bharadwaj, & Grover, 2003), digital infrastructure evolution (Tilson et al., 2010; Henfridsson & Bygstad, 2013), and digital innovation management (Yoo, Henfridsson, & Lyytinen, 2010; Nambisan et al., 2017; Kohli & Melville, 2019), this study has articulated a multi-stage activation model. The model explains how Knowledge Management Systems (KMS) transform dormant data through identification, structuring, contextualization, integration, and recombination. Each stage contributes to cumulative capability development, ultimately enhancing organizational agility, innovation capacity, governance quality, and performance outcomes.

The central contribution of this work lies in reframing KMS as dynamic capability infrastructures rather than static repositories. When effectively designed and strategically aligned, KMS mediate between heterogeneous digital infrastructures and organizational decision processes. They enable interoperability, reduce interpretive ambiguity, and expand the portfolio of recombinable knowledge components within digital ecosystems. In doing so, they convert latent informational residues into strategic digital options that can be mobilized during periods of environmental uncertainty or technological change.

Moreover, the activation perspective broadens the conversation on digital innovation by highlighting the importance of historical and archival data in shaping future value creation. Innovation is not only driven by new data acquisition but also by intelligent reuse and reinterpretation of accumulated knowledge assets. Dormant data, once activated, becomes a generative input into service transformation, operational optimization, and strategic renewal.

Future research should empirically test the proposed activation pathway and explore contextual moderators such as industry digital maturity, governance structures, and cultural readiness for knowledge sharing. Nonetheless, the conceptual framework developed here provides a robust foundation for understanding how organizations can systematically unlock the hidden value embedded within their own digital archives. In an era defined by data abundance, sustainable competitive advantage may increasingly depend not on collecting more data, but on activating what organizations already possess.

References

1. Sambamurthy, V., Bharadwaj, A., & Grover, V. (2003). Shaping agility through digital options. MIS Quarterly, 27(2), 237–263. https://doi.org/10.2307/30036530

2. Rai, A., Patnayakuni, R., & Seth, N. (2006). Firm performance impacts of digitally enabled supply chains. MIS Quarterly, 30(2), 225–246. https://doi.org/10.2307/25148729

3. Olaitan, O. F., Adebanjo, T. A., Obozokhai, L. I., Iwerumoh, A. N., Balogun, I. O., & Ojo, D. A. (2025). Dark data in business intelligence: A systematic review of challenges, opportunities, and value creation potential. Journal of Economics, Business, and Commerce, 2(2), 135-142. https://doi.org/10.69739/jebc.v2i2.1000

4. Melville, N., Kraemer, K., & Gurbaxani, V. (2004). IT and organizational performance. MIS Quarterly, 28(2), 283–322. https://doi.org/10.2307/25148636

5. Aral, S., & Weill, P. (2007). IT assets, organizational capabilities, and firm performance. Organization Science, 18(5), 763–780. https://doi.org/10.1287/orsc.1070.0308

6. Yoo, Y., Henfridsson, O., & Lyytinen, K. (2010). Research commentary—The new organizing logic of digital innovation. Information Systems Research, 21(4), 724–735. https://doi.org/10.1287/isre.1100.0322

7. Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital innovation management. MIS Quarterly, 41(1), 223–238. https://doi.org/10.25300/MISQ/2017/41:1.03

8. Henfridsson, O., & Bygstad, B. (2013). The generative mechanisms of digital infrastructure evolution. MIS Quarterly, 37(3), 907–931. https://doi.org/10.25300/MISQ/2013/37.3.11

9. Barrett, M., et al. (2015). Service innovation in the digital age. MIS Quarterly, 39(1), 135–154. https://doi.org/10.25300/MISQ/2015/39.1.07

10. Tilson, D., et al. (2010). Digital infrastructures. Information Systems Research, 21(4), 748–759. https://doi.org/10.1287/isre.1100.0328

11. Kohli, R., & Melville, N. (2019). Digital innovation: A review and synthesis. Information Systems Journal, 29(1), 200–223. https://doi.org/10.1111/isj.12193